

NZ AI Policy Sandbox – Model Equations v0.1

Tier 1 Sector Dynamics: Bounded Gain-Loss ODE System

New Zealand Whole-Economy Policy Sandbox

AI for Good New Zealand

Version 0.1 – April 2026

Abstract

This document specifies the mathematical foundations of the NZ AI Policy Sandbox (NZAIS) for Tier 1 sectors. The sandbox is a sector-calibrated, whole-economy model for comparing AI policy approaches in New Zealand. It is designed for policy comparison, not GDP forecasting.

The model adapts the bounded gain-loss ODE structure from Strauss [1] to a 19-sector tiered economy. Each Tier 1 sector carries four state variables: absorptive capability (K_s), AI adoption maturity (A_s), realised productivity effect (P_s), and labour adjustment pressure (L_s). A single economy-wide enabling stock (E) supports all sectors via the national infrastructure layer.

Design principles from `STATE-VARIABLES.md`: one mechanism per layer (no double counting), bounded state variables by construction, sector blocks as the unit of analysis, whole-economy denominator honesty, and comparative usefulness over predictive theatre.

This v0.1 document specifies the *form* of the equations for Tier 1 only. Parameter calibration is deferred to v0.2. Tier 2 and Tier 3 sectors are deferred to v0.2 and v0.3 respectively.

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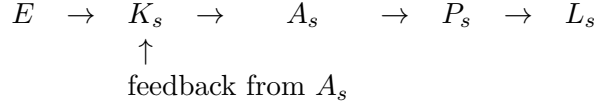
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1 Model Structure

1.1 Design Principle: One Mechanism Per Layer

The model follows a strict causal chain where each variable is responsible for one layer. Each arrow in the DAG below represents exactly one mechanism and appears nowhere else in the system.



- E does not act directly on adoption - it acts *through* K_s (enabling capacity raises absorptive capability).
- K_s does not create productivity directly - it enables adoption *through* A_s .
- A_s does not create labour pressure directly - it creates productivity *through* P_s , and labour pressure emerges *through* L_s .
- P_s is driven by adoption; L_s is driven by adoption and sector structure (not by P_s directly, to avoid double counting).
- There is one feedback: adoption A_s reinforces the demand for capability K_s (learning-by-doing, procurement maturity).

No mechanism appears twice. Every causal path is traceable through a single logical chain.

1.2 State Variables

Definition 1 (Tier 1 Sector State Variables). *For each sector $s \in \mathcal{S}_1$ (Tier 1, $|\mathcal{S}_1| = 9$), the model tracks four state variables, all bounded in $[0, 1]$:*

$$K_s(t) : \text{Absorptive capability} - \text{effective capacity to absorb AI into productive use} \quad (1)$$

$$A_s(t) : \text{AI adoption maturity} - \text{degree of meaningful operational AI deployment} \quad (2)$$

$$P_s(t) : \text{Realised productivity effect} - \text{normalised realisation index, bounded in } [0, 1] \quad (3)$$

$$L_s(t) : \text{Labour adjustment pressure} - \text{displacement and reconfiguration index} \quad (4)$$

Definition 2 (Economy-Wide State Variable). *There is one economy-wide enabling state:*

$$E(t) : \text{National enabling capacity} - \text{economy-wide environment for AI adoption} \quad (5)$$

bounded in $[0, 1]$.

Remark 1. $P_s(t) \in [0, 1]$ by normalisation to the sector-specific productivity ceiling $\bar{p}_s$. The raw productivity gain is $\bar{p}_s \cdot P_s(t)$, where $\bar{p}_s$ is a sector parameter calibrated separately (v0.2). In v0.1, only the normalised form is specified.

Remark 2. The GDP share ω_s for each sector is a constant input parameter, not a state variable. It is sourced from `data/sector-parameter-table.csv` (Stats NZ GDP by Industry). See Section 4.

Tier 1 Sectors

The nine Tier 1 sectors are:

Index	Sector	Symbol
1	Agriculture	$s = \text{ag}$
2	Manufacturing	$s = \text{mfg}$
3	Professional Services	$s = \text{prof}$
4	Public Sector	$s = \text{pub}$
5	Technology	$s = \text{tech}$
6	Healthcare	$s = \text{hlth}$
7	Construction	$s = \text{con}$
8	Financial Services	$s = \text{fin}$
9	Retail & Wholesale	$s = \text{ret}$

2 Core Dynamics

Assumption 1 (Nonnegative Rates). *All rate parameters are nonnegative:*

$$\rho_K, \rho_A, \rho_P, \rho_L, \rho_E, \alpha_s, \kappa_s, \phi_s, \lambda_s, \eta_s, \theta_s \geq 0.$$

This ensures the bounded gain-loss forms keep all state variables in $[0, 1]$ by construction, without requiring ex-post clamping.

All equations use bounded gain-loss forms:

$$\frac{dX}{dt} = (1 - X) \cdot \rho_X \cdot f(\text{drivers}) - X \cdot \rho_X \cdot g(\text{drivers})$$

so $X \in [0, 1]$ is maintained for all $t \geq 0$. See Section 5 for the formal proposition.

2.1 Capability Stock Dynamics – $K_s(t)$

Absorptive capability grows when the national enabling environment is strong and when sector-specific support is provided. It decays in the absence of investment and maintenance.

$$\boxed{\frac{dK_s}{dt} = (1 - K_s) \cdot \rho_K \cdot (\phi_s E + G_s) - K_s \cdot \rho_K \cdot \mu_s} \quad (6)$$

Interpretation:

- $(1 - K_s) \cdot \rho_K \cdot \phi_s E$: national enabling capacity lifts sector capability, at rate proportional to the sector's exposure ϕ_s to the national environment.
- $(1 - K_s) \cdot \rho_K \cdot G_s$: sector-specific policy support G_s directly builds capability.
- $K_s \cdot \rho_K \cdot \mu_s$: capability decays at rate μ_s (skill attrition, technology obsolescence, institutional drift).
- $(1 - K_s)$ saturation prevents $K_s > 1$; K_s decay prevents $K_s < 0$.

Feedback from adoption: Adoption-driven demand reinforces capability building. This is modelled as an additive term in the gain:

$$\boxed{\frac{dK_s}{dt} = (1 - K_s) \cdot \rho_K \cdot (\phi_s E + G_s + \eta_s A_s) - K_s \cdot \rho_K \cdot \mu_s} \quad (7)$$

where $\eta_s \geq 0$ is the adoption feedback parameter: as adoption grows, demand for capability (training, procurement maturity, data infrastructure) reinforces the capability stock. This is the one feedback in the DAG; it does not appear elsewhere.

Remark 3. G_s is a policy control input (exogenous), not a state variable. In the base scenario (no intervention), $G_s = 0$.

2.2 AI Adoption Dynamics – $A_s(t)$

Adoption grows when capability is high and decays in the absence of enabling conditions.

$$\boxed{\frac{dA_s}{dt} = (1 - A_s) \cdot \rho_A \cdot \alpha_s K_s - A_s \cdot \rho_A \cdot (1 - K_s)} \quad (8)$$

Interpretation:

- $(1 - A_s) \cdot \rho_A \cdot \alpha_s K_s$: adoption expands when capability is available. $\alpha_s \geq 0$ is the adoption responsiveness to capability for sector s .
- $A_s \cdot \rho_A \cdot (1 - K_s)$: when capability is weak, existing adoption stalls or reverses (firms de-adopt, fail to maintain, or substitute away). The $(1 - K_s)$ driver ensures that low capability creates adoption pressure loss.

Remark 4. E does not appear directly in dA_s/dt . The national enabling environment acts only through K_s . This preserves the one-mechanism-per-layer principle.

2.3 Productivity Dynamics – $P_s(t)$

Realised productivity grows as adoption matures. It decays slowly when adoption is low (reversal, skill loss, workflow regression).

$$\boxed{\frac{dP_s}{dt} = (1 - P_s) \cdot \rho_P \cdot \kappa_s A_s - P_s \cdot \rho_P \cdot (1 - A_s)} \quad (9)$$

Interpretation:

- $(1 - P_s) \cdot \rho_P \cdot \kappa_s A_s$: realised productivity gains accumulate with adoption. $\kappa_s \geq 0$ is the productivity responsiveness to adoption for sector s .
- $P_s \cdot \rho_P \cdot (1 - A_s)$: when adoption falls, the realised gain decays back toward zero at rate proportional to $(1 - A_s)$.

Remark 5. K_s does not appear directly in dP_s/dt . Capability affects productivity only through A_s . This is the one-mechanism-per-layer constraint.

2.4 Labour Adjustment Pressure – $L_s(t)$

Labour adjustment pressure rises as adoption expands, modulated by sector-specific labour sensitivity. It eases when adoption levels off.

$$\boxed{\frac{dL_s}{dt} = (1 - L_s) \cdot \rho_L \cdot \lambda_s A_s - L_s \cdot \rho_L \cdot (1 - A_s)} \quad (10)$$

Interpretation:

- $(1 - L_s) \cdot \rho_L \cdot \lambda_s A_s$: pressure rises with adoption. $\lambda_s \geq 0$ captures sector-specific labour sensitivity (displacement exposure, role reconfiguration, skills mismatch risk).

- $L_s \cdot \rho_L \cdot (1 - A_s)$: pressure eases as adoption plateaus or reverses (labour market adjusts, redeployment occurs, skills rebalance).

Remark 6. P_s does not appear in dL_s/dt . Labour pressure is driven by adoption, not directly by productivity gains - consistent with the design principle that adoption and productivity are causally separated. This avoids double counting.

2.5 National Enabling Capacity – $E(t)$

$E(t)$ is the single economy-wide state. It grows with national investment G_E and decays with depreciation.

$$\boxed{\frac{dE}{dt} = (1 - E) \cdot \rho_E \cdot G_E - E \cdot \rho_E \cdot \delta_E} \quad (11)$$

Interpretation:

- $(1 - E) \cdot \rho_E \cdot G_E$: national enabling investment G_E lifts the economy-wide enabling stock (skills pipeline, digital public infrastructure, diffusion mechanisms).
- $E \cdot \rho_E \cdot \delta_E$: the enabling environment depreciates at rate δ_E (skills attrition, infrastructure ageing, policy reversal).

Remark 7. E acts on the whole economy simultaneously. In the base scenario (no supply-side intervention), $G_E = G_E^{\text{base}} \geq 0$, representing existing background enabling investment (e.g., existing government digital programmes).

3 Policy Scenarios

All three scenarios share the same equal budget $\Delta \geq 0$:

$$\Delta_A = \Delta_B = \Delta_C = \Delta.$$

This is required for like-for-like comparison. Let $n = |\mathcal{S}_1| = 9$.

3.1 Scenario A – Aggregate Policy

Policy support is distributed uniformly across all Tier 1 sectors by equal per-sector allocation.

Perturbation: Add Δ_A/n to the capability gain term for every sector $s \in \mathcal{S}_1$:

$$G_s^{(A)} = \frac{\Delta_A}{n}, \quad \forall s \in \mathcal{S}_1. \quad (12)$$

The capability equation under Scenario A becomes:

$$\frac{dK_s}{dt} = (1 - K_s) \cdot \rho_K \cdot \left(\phi_s E + \frac{\Delta_A}{n} + \eta_s A_s \right) - K_s \cdot \rho_K \cdot \mu_s.$$

3.2 Scenario B – Targeted Demand-Side Policy

Policy support is weighted toward sectors with greater adoption gaps, reflecting demand-side targeting.

Perturbation: Add $\Delta_B \cdot w_s^{\text{demand}}$ to the capability gain term, where the demand weight is:

$$w_s^{\text{demand}} = \frac{1 - A_s(0)}{\sum_{s' \in \mathcal{S}_1} (1 - A_{s'}(0))}, \quad \forall s \in \mathcal{S}_1. \quad (13)$$

so that $\sum_s w_s^{\text{demand}} = 1$ and the total budget is exactly Δ_B .

The capability equation under Scenario B becomes:

$$\frac{dK_s}{dt} = (1 - K_s) \cdot \rho_K \cdot \left(\phi_s E + \Delta_B \cdot w_s^{\text{demand}} + \eta_s A_s \right) - K_s \cdot \rho_K \cdot \mu_s.$$

Remark 8. w_s^{demand} is evaluated at the initial condition $A_s(0)$ and held constant over the scenario horizon. This represents a one-off policy allocation decision, not a continuously adaptive rule.

3.3 Scenario C – Targeted Supply-Side Policy

Policy support is concentrated entirely on the national enabling stock, on the theory that lifting E creates spillover benefits across all sectors simultaneously.

Perturbation: Add Δ_C to the E gain term. No direct sector capability support.

$$G_E^{(C)} = G_E^{\text{base}} + \Delta_C, \quad G_s^{(C)} = 0 \quad \forall s \in \mathcal{S}_1. \quad (14)$$

The enabling stock equation under Scenario C becomes:

$$\frac{dE}{dt} = (1 - E) \cdot \rho_E \cdot (G_E^{\text{base}} + \Delta_C) - E \cdot \rho_E \cdot \delta_E.$$

Remark 9. Scenario C does not directly fund any sector. It invests in the enabling environment - skills pipeline, digital public infrastructure, diffusion mechanisms. Sector capability responds to E via the $\phi_s E$ term in Equation (7). Sectors with higher ϕ_s benefit more from supply-side investment.

Remark 10. The central analytical question - aggregate versus targeted, and direct support versus enabling investment - is precisely the comparison between Scenarios A, B, and C under the equal budget constraint $\Delta_A = \Delta_B = \Delta_C = \Delta$.

4 Aggregation

4.1 Tier 1 Productivity Aggregate

The GDP-weighted Tier 1 productivity aggregate is:

$$\bar{P}_{T1}(t) = \sum_{s \in \mathcal{S}_1} \omega_s P_s(t) \quad (15)$$

where ω_s is sector s 's share of total GDP (sourced from `data/sector-parameter-table.csv`, Stats NZ GDP by Industry Dec 2025).

Whole-economy denominator honesty: The Tier 1 sectors cover approximately 61% of New Zealand GDP. $\bar{P}_{T1}(t)$ is therefore *not* a full-economy productivity aggregate. It is an aggregate only over the nine Tier 1 sectors modelled here.

The full whole-economy aggregate will be:

$$\bar{P}(t) = \sum_{s \in \mathcal{S}} \omega_s P_s(t) \quad (16)$$

where $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2 \cup \mathcal{S}_3$ spans all 19 sectors. This requires Tier 2 (v0.2) and Tier 3 (v0.3).

4.2 GDP Share Parameters

Sector GDP shares ω_s are normalised such that $\sum_{s \in \mathcal{S}} \omega_s = 1$ across all 19 sectors. Within Tier 1 only, the shares are re-normalised for the partial aggregate $\bar{P}_{T1}(t)$ - but the v0.1 document preserves the raw ω_s from the parameter table so that Tier 2/3 extension is additive, not a re-weighting exercise.

5 Boundedness

Proposition 1 (State Variables Remain Bounded). *If all state variables satisfy their initial conditions $K_s(0), A_s(0), P_s(0), L_s(0), E(0) \in [0, 1]$, and all rate parameters satisfy Assumption 1, then for all $t \geq 0$:*

$$K_s(t), A_s(t), P_s(t), L_s(t), E(t) \in [0, 1].$$

Proof. Each equation is of the bounded gain-loss form $\dot{X} = (1 - X) \cdot f^+(X) - X \cdot f^-(X)$ where $f^+, f^- \geq 0$ by Assumption 1. When $X = 1$, the gain term vanishes and the loss term is $-f^-(1) \leq 0$, so $\dot{X} \leq 0$ at $X = 1$: the variable cannot increase above 1. When $X = 0$, the loss term vanishes and the gain term is $(1) \cdot f^+(0) \geq 0$, so $\dot{X} \geq 0$ at $X = 0$: the variable cannot decrease below 0. Together, $\{[0, 1]\}$ is a positively invariant set for each equation. Since the equations are coupled through bounded terms, and each individual state variable cannot leave $[0, 1]$, the joint state space $[0, 1]^{4n+1}$ (for n Tier 1 sectors plus E) is positively invariant. $\square$

6 Sign Correctness

Each partial derivative should have an economically sensible sign. The table below summarises the key signs for the Tier 1 dynamics.

Equation	Partial derivative	Sign	Interpretation
dK_s/dt	$\partial/\partial E$	+	More enabling capacity lifts capability
dK_s/dt	$\partial/\partial G_s$	+	More sector support builds capability
dK_s/dt	$\partial/\partial A_s$	+	Adoption reinforces capability demand
dK_s/dt	$\partial/\partial K_s$	− (net)	Saturation and decay dominate at high K_s
dA_s/dt	$\partial/\partial K_s$	+	Higher capability enables more adoption
dA_s/dt	$\partial/\partial A_s$	− (net)	Saturation dominates at high A_s
dP_s/dt	$\partial/\partial A_s$	+	More adoption realises more productivity
dP_s/dt	$\partial/\partial P_s$	− (net)	Saturation dominates at high P_s
dL_s/dt	$\partial/\partial A_s$	+	More adoption raises labour pressure
dL_s/dt	$\partial/\partial L_s$	− (net)	Saturation: pressure cannot exceed 1
dE/dt	$\partial/\partial G_E$	+	More national investment lifts enabling stock
dE/dt	$\partial/\partial E$	− (net)	Saturation and depreciation dominate at high E

All signs are economically sensible and consistent with the causal DAG.

7 Calibration Pointers

Version 0.1 specifies the *form* of the equations. Numerical parameter values are deferred to v0.2. This section notes what each parameter class requires.

Source: All baseline parameter values will be drawn from `data/sector-parameter-table.csv` (348-cell parameter table, 35% pre-populated as of v0.1 release). Evidence quality is classified as observed, derived, or assumed per the data philosophy in `METHODS.md`.

Parameter	Description	Evidence class
ω_s	Sector GDP share	Observed (Stats NZ)
$A_s(0)$	Initial adoption state	Derived / assumed (sector research)
$K_s(0)$	Initial capability state	Assumed (digital maturity proxies)
$P_s(0)$	Initial productivity state	Assumed (sector productivity evidence)
$L_s(0)$	Initial labour pressure	Assumed (automation exposure logic)
$E(0)$	Initial enabling stock	Assumed (NZ digital infrastructure signals)
$\rho_K, \rho_A, \rho_P, \rho_L, \rho_E$	Adjustment speeds	Assumed (tuned for plausibility)
α_s	Adoption responsiveness to K_s	Derived (sector archetype)
κ_s	Productivity responsiveness	Derived (OECD productivity range)
λ_s	Labour sensitivity	Derived (automation exposure)
ϕ_s	Enabling exposure	Assumed (sector digital openness)
η_s	Adoption feedback to K_s	Assumed (learning-by-doing proxy)
μ_s	Capability decay rate	Assumed
δ_E	Enabling stock decay	Assumed

Remark 11. *Do not interpret any specific numerical values in v0.1 as calibrated. Parameter values shown in any illustrative computations are placeholders. The v0.2 calibration note will provide sector-specific fitted values anchored to the evidence in `data/sector-parameter-table.csv`.*

8 Known Limits of v0.1

This document is explicit about what it does and does not claim.

1. **Tier 1 only – partial economy coverage.** The nine Tier 1 sectors modelled here cover approximately 61% of New Zealand GDP and approximately 70% of employment. The Tier 1 aggregate $\bar{P}_{T1}(t)$ is not a whole-economy aggregate. Tiers 2 and 3 are deferred to v0.2 and v0.3.
2. **Parameter values not calibrated.** Only the functional form of each equation is specified in v0.1. No sector-specific parameter values have been fitted to data. Calibration is v0.2 work.
3. **Deterministic ODE only.** This is a deterministic, continuous-time ODE system. No stochastic shocks, no uncertainty propagation, no Monte Carlo. Stochastic extensions are deferred to a later version.
4. **Single-country model.** The model represents New Zealand only. There are no trade terms, international spillovers, or cross-border adoption dynamics. NZ’s small-open-economy characteristics are not yet modelled.
5. **No sector-pair interactions.** Each sector block in v0.1 is diagonal - sectors interact only through the shared enabling stock $E(t)$. Direct sector-pair linkages (e.g., Technology enabling Manufacturing; Financial Services intermediating Construction) are not yet in the system. This is the principal structural simplification in v0.1.
6. **Labour pressure is a reduced-form index.** $L_s(t)$ is not a labour market model. It does not produce job counts, wage forecasts, or distributional effects. It is a bounded pressure index designed to capture the direction and relative severity of labour adjustment, not its precise magnitude.
7. **Policy controls are constant within a scenario.** G_s and G_E are treated as fixed scenario parameters, not adaptive policies that respond to system state. An adaptive or feedback policy formulation is deferred.

9 References

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